

Adriahna Blackburn

www.linkedin.com/in/adriahna-blackburn

Education

Oklahoma State University (OSU), Stillwater, OK

May 2026

B.S. in Microbiology/Cell & Molecular Biology, Honors

Honors College thesis title: "Pyocin pathway mysteries: exploring the role of two novel proteins for pyocin production in *Pseudomonas aeruginosa*"

Work/Research Experience

Fleming Research Scholar

June 2025-July 2025

Oklahoma Medical Research Foundation (Dawson Lab)—Oklahoma City, OK

- Elucidated the molecular production pathway of bacteriocins produced by the bacterium *Pseudomonas Aeruginosa*
- 1 of 14 scholars chosen for an 8-week biomedical research internship
- Investigated how certain physical properties of chromosomes correlate to chromosome segregation errors during meiosis
- Improved microscopy measuring criteria of yeast chromosomes
- Utilized epi-fluorescent microscopy techniques, yeast plasmid transformation, and data analysis
- Improved scientific communication by presenting work to a layman audience via an oral presentation

Beckman Research Scholar/Undergraduate Researcher

May 2023-May 2025

Oklahoma State University Department of Microbiology & Molecular Genetics (Cabeen Lab)—Stillwater, OK

- Elucidated the molecular production pathway of bacteriocins called pyocins made by the bacterium *Pseudomonas Aeruginosa*
- Improved scientific communication for a broad audience and was selected as 1 of 4 national Beckman Scholars to give an oral presentation at the annual Beckman Symposium
- Mastered molecular cloning techniques, DNA/RNA extractions, PCR, procedure optimization, and critical thinking

Peer Tutor

August 2023-December 2023

Oklahoma State University—Stillwater, OK

- Guided tutoring sessions for Introduction to Biology, Introduction to Microbiology, and Inorganic Chemistry I (1-hour long sessions for 8 hours a week)
- Developed communication skills, engaged others in active learning, and learned how to cater knowledge to a wide array of students

Extra Curriculars

Project Co-Leader | OSU International Genetically Engineered Machine (iGEM)

May 2024-May 2025

Oklahoma State University—Stillwater, OK

- Developed and led a synthetic biology project to present at the international competition in Paris, France
- Researched applications of photodynamic therapy, bacteriophages, and reactive oxygen species to treat *Staphylococcus aureus* skin infections
- Taught other club members molecular cloning and basic microbiology laboratory techniques
- Led outreach efforts at local middle schools by running science experiments such as DNA extractions

President | OSU Society of Asian Scientists and Engineers (SASE)

May 2024-May 2025

Oklahoma State University—Stillwater, OK

- Prepared Asian heritage scientists and engineers to be successful in the business world and in their careers by organizing professional workshops, company presentations, attendance at the SASE National Conference, social events, volunteering, and cultural celebrations
- Oversaw E-board activities and responsibilities and acted as a point of accountability

Executive Member | OSU College of Arts and Sciences Research Leaders

May 2023-May 2025

Oklahoma State University—Stillwater, OK

- Planned and executed workshops to help other undergraduate students navigate research experiences at OSU
- Conducted classroom and club visits to provide information about my experiences in undergraduate research and information about the Beckman Scholars Program

Honors and Awards

- NSF GRFP Honorable Mention Spring 2026
- Goldwater Scholar Spring 2025
- Beckman Scholars Program- 1 of 2 inaugurating Beckman Scholars at OSU Spring 2023-Fall 2024
- American Society of Microbiology Missouri Valley Branch - Best Undergraduate Poster Presentation, 1st Spring 2024
- OSU College of Arts and Sciences Community Service Leader Award Spring 2023